

Response to Reviewers — Second Round

Manuscript: “Lifetime Performance and Degradation of Floating Photovoltaic Systems Under Humid and Saline Conditions”

I thank the reviewer for the second careful assessment and for the constructive view of the first revision. This round addresses the remaining concerns as follows:

1. Baseline and Monte Carlo results are now distinguished explicitly in the abstract, results and conclusion.
2. The conditional nature of the freshwater benefit is now a headline conclusion.
3. A full Monte Carlo parameter table with sources has been added (Appendix, Table XIII).
4. Financing uncertainty is included (discount rate and O&M escalation, Table XI and the Monte Carlo).
5. The constant-rate lifetime projection is stated explicitly.
6. The supported physics and the assumed functional forms of the salt–PID coupling are separated.
7. A concrete field-validation protocol has been added (Table XII).

The two financing parameters raise the Monte Carlo to 28 parameters. Because the random sequence therefore changes, all Monte Carlo statistics are regenerated. They differ from the first revision only within sampling noise, for example freshwater median +1.5% instead of +1.3%, and saline median –5.1% instead of –4.9%. The baseline results are unchanged. As before, every number is generated by one script.

Comment 1 (§4, mitigation wording). *Call the mitigation results “scenario-based mitigation benefits”, not validated benefits.*

Response. Done. Table VIII is retitled “Scenario-Based Mitigation Benefits for the Saline FPV Plant”, and Section IV-E states that these are scenario-based mitigation benefits, not validated ones. The “what-if” language is retained.

Comment 2 (§5, thermal headline). *Make the dependence of the freshwater benefit on the thermal characteristics of the floating structure a headline conclusion.*

Response. Done. The abstract now states that “the freshwater benefit is conditional: it is positive in 75% of samples, depends strongly on the thermal characteristics of the floating structure, and vanishes if the dew point over the water rises by more than about 2.7 K.” The conclusion repeats this, and the new Section V-B states as a practical rule that the benefit “should not be assumed from the presence of water alone”.

Comment 3 (§6, weather wording). *Use “climatologically consistent synthetic weather generator” rather than “validated”.*

Response. Done. Section III-A now says the generator “is climatologically consistent with its inputs; it is *not* an external validation”. The word “validated” is not used for the weather model anywhere in the manuscript.

Comment 4 (§7 and §22.6, field validation). *Add a validation case or, if unavailable, present the work as a scenario and uncertainty framework requiring field calibration, with a concrete validation protocol.*

Response. Measured FPV data are still not available to this study, so the second option is taken. New Section V-E and Table XII propose a minimum measurement set for an operating FPV plant with a co-located land reference:

- GHI and POA irradiance;
- air temperature, RH and wind over water and over land;
- water temperature;

- module temperature at several positions;
- a dew/wetness sensor on the glass;
- cleaned and uncleaned soiling reference cells;
- chloride deposition by wet candle (ISO 9225);
- rainfall and string DC power;
- insulation resistance and leakage current;
- quarterly IV curves of reference modules.

Each measurement is mapped to the model parameter it would calibrate. The abstract and conclusion keep the statement that the results are “model-based scenario predictions that still require field validation”.

Changes: Section V-E and Table XII (new).

Comment 5 (§9 and §22.5, constant degradation rate). *State that the lifetime projection assumes a constant effective annual degradation rate.*

Response. Added to Section III-H, essentially in the reviewer’s words: “The lifetime projection assumes a constant effective annual degradation rate derived from the stress distribution of the simulated year; the 25-year results therefore represent an equivalent-rate scenario, not a mechanistic year-by-year degradation trajectory.”

Comment 6 (§10, salt–PID coupling). *Distinguish the experimentally supported physics from the assumed leakage function and threshold shift.*

Response. Done. Section III-F now separates *experimentally supported physics* (PID leakage rises steeply with humidity; sea salt lowers glass surface resistance by orders of magnitude and raises leakage current) from the *assumed model representation* (the logistic form, its width, RH_{50} and $\Delta_s = 15\%$ RH, none of which has been fitted to data).

Comment 7 (§11, soiling). *State that the absolute soiling losses are scenario estimates.*

Response. Added to Section IV-B: “The absolute soiling losses are scenario estimates, because the salt and dust deposition rates were not measured at the reference site.”

Comment 8 (§13 and §22.4, economic uncertainty). *State that the economic conclusions are location- and year-dependent; add sensitivity to the discount rate (6–12%) and O&M escalation (2–7%).*

Response. Both are done:

- New Table XI gives the LCOEs of the three plants for discount rates of 6, 9 and 12% and O&M escalation of 2, 5 and 7%/yr. The financing assumptions shift all LCOEs but not their ranking: the gap to land PV ranges from 0.58 to 0.86 INR/kWh for freshwater FPV and from 1.36 to 1.87 INR/kWh for saline FPV.
- Both parameters are now sampled in the Monte Carlo. Freshwater FPV is more expensive than land PV in 99.5% of samples, and saline FPV in 100.0%.
- Sections III-H and IV-I state that the economic results are location- and year-dependent and specific to the stated Indian-market assumptions.

Changes: Sections III-H, III-I, IV-I; Table XI (new); Table XIII.

Comment 9 (§14 and §22.2, baseline vs. Monte Carlo). *Label baseline and Monte Carlo results consistently, e.g. saline baseline −4.3% vs. median −4.94%.*

Response. Done throughout. The abstract now reports the baseline differences and then the Monte Carlo medians with their 95% intervals. Section IV-H opens with: “Baseline values and Monte Carlo medians answer different questions and are reported separately.” It continues: “For freshwater FPV, the baseline 25-year difference is +2.1%, while the Monte Carlo median is +1.5% with a 95% interval of [−3.6, 6.3]%. The saline and mitigated cases follow in the same form. The

section also explains why medians and baselines differ: several ranges are asymmetric about the baseline, notably the FPV heat-loss coefficients. The conclusion uses the same two-part structure.

Comment 10 (§15–16, interpretation). *Emphasize that the freshwater advantage is conditional rather than robust; phrase the saline result as “the model predicts a lifetime penalty for saline FPV across most of the plausible parameter space”.*

Response. Both phrasings are adopted verbatim in the abstract, Section IV-H and the conclusion.

Comment 11 (§17, ΔT_d^*). *Turn the 2.7 K threshold into a practical recommendation.*

Response. Section V-B now states: “the dew-point (humidity) increment over the water should be measured before a lifetime benefit is claimed, because the benefit vanishes above about 2.7 K.” The final paragraph of the conclusion repeats it as a recommendation for FPV assessments.

Comment 12 (§18, Fig. 12 in the Discussion). *Refer to the Monte Carlo/Spearman analysis more prominently in the Discussion.*

Response. New Section V-B (“Decision-Relevant Drivers”) builds on Fig. 12(b). It identifies which inputs decide each comparison and should therefore be measured first at a candidate site. The validation protocol (Section V-E) refers back to it.

Comment 13 (§21, novelty claim). *Soften “To the author’s knowledge, no study yet resolves. . .”.*

Response. Changed to: “Among the studies reviewed here, none was identified that resolves, within one consistent microclimate, . . .”.

Comment 14 (§22.3, Monte Carlo parameter table). *Provide a table with parameter, baseline, distribution, bounds and justification/reference.*

Response. Added as Appendix A, Table XIII, covering all 28 parameters. The table is generated from the same parameter definitions that drive the Monte Carlo (`fpv_analysis/uncertainty.py`), so it cannot drift from the code. The justifications cite measured FPV heat-loss coefficients, ISO 9223 chloride classes and open-rack defaults where available. All other entries are labelled “Assumption” or “Calibration constant” where no source exists.

Changes: Appendix A and Table XIII (new); Section III-I.

Comment 15 (§22.1, model vs. validation wording). *Use “model prediction” rather than “performance” where appropriate.*

Response. The abstract and conclusion now say “the model predicts” throughout. Results remain prefaced by the statement that all results are model predictions under the baseline assumptions.

Comment 16 (§23, minor issues). *Year-1 vs. year 1; notation; symbol definitions; salt vs. chloride deposition; units; capitalization; references.*

Response.

- “Year-1” is used as an adjective (“year-1 yield”) and “year 1” as a noun, consistently.
- R , R_{TH} , R_{PID} and R_C are typeset uniformly. The superscript “ref” denotes the reference rates.
- Section III-D now defines the two salt quantities explicitly. The *salt deposition* r_s is the sea-salt mass retained on the glass (g/m²/day), which drives soiling and leakage. The *chloride deposition* D_{Cl} is the chloride-ion deposition rate of the atmosphere in the ISO 9223 sense (mg/m²/day), which drives corrosion.
- Units of deposition are given consistently in the text and tables.
- “FPV” is capitalized throughout.
- References use the IEEEtran style. Volume, page and DOI details for the references added in the first revision are being completed from the publisher records.

Remaining limitation

The absence of measured FPV and site-weather data remains the main limitation. The manuscript now presents the work as a physically informed scenario and uncertainty framework that requires field calibration, and it proposes the measurement protocol needed to do so.